

Winfree Oscillatory Neural Network

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Abstract

Oscillations and synchronization are widely believed to play a fundamental role in representation and computation. However, existing machine learning approaches based on synchronization dynamics have largely been confined to specialized settings such as object discovery, with limited evidence of scalability to standard vision benchmarks or logic reasoning tasks. We propose the *Winfree Oscillatory Neural Network* (*WONN*), a dynamical neural architecture based on generalized Winfree dynamics. *WONN* evolves representations on the torus $(S^1)^d$ through structured oscillatory interactions, combining phase-based inductive biases with flexible and hierarchical interaction mechanisms instantiated as either fixed trigonometric mappings or learnable neural networks. We evaluate *WONN* on image recognition and complex reasoning tasks, including CIFAR, ImageNet, Maze-hard, and Sudoku. Across these domains, *WONN* achieves competitive or superior performance with strong parameter efficiency. **In particular, *WONN* is, to our knowledge, the first synchronization-based oscillatory architecture to scale competitively to ImageNet-1K.** Furthermore, on Maze-hard, *WONN* achieves **80.1% accuracy using only 1% of the parameters of prior state-of-the-art models**. These results suggest that structured oscillatory dynamics provide a scalable and parameter-efficient alternative to conventional neural architectures. The project page is available at https://jiawen-dai.github.io/WONN_Project_Page/.

1 Introduction

Oscillations and synchronization are ubiquitous mechanisms for organizing collective dynamics in biological and physical systems [2, 46, 20]. In neuroscience, oscillatory activity has long been associated with temporal coordination, inter-area communication, and structured information processing [2, 8, 27], while synchronization and temporal correlation have been proposed as mechanisms for perceptual binding and coherent representation formation [42, 43, 34]. In physics, coupled oscillator systems provide canonical examples of how simple local interactions can self-organize into coherent macroscopic behavior [46, 20]. Together, these observations suggest an alternative perspective on neural computation: rather than viewing representation learning purely as a sequence of static feature

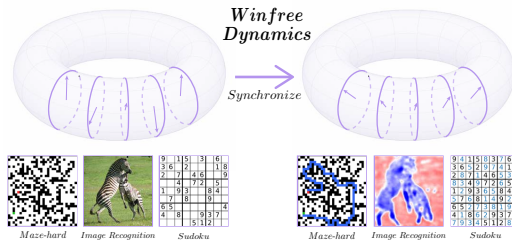


Figure 1: Illustration of the proposed *WONN*. The network evolves on the toroidal phase space $(S^1)^d$, where neurons are represented as phase oscillators. Through generalized Winfree synchronization dynamics, oscillators self-organize into structured collective states, enabling effective computation for both image recognition and reasoning tasks.

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transformations, computation may instead emerge from the collective evolution and synchronization of interacting dynamical states.

Despite their conceptual appeal, synchronization-based principles have so far had limited impact on mainstream machine learning. Existing synchronization-inspired approaches are primarily restricted to specialized settings such as object-centric representation learning or unsupervised object discovery [24, 39, 10, 23]. More recent architectures such as *AKOrN* [26] introduce Kuramoto-type oscillatory neurons into neural networks, but the empirical scalability of oscillatory architectures remains largely unclear. In particular, it is unknown whether synchronization-based computation can scale to large-scale image recognition benchmarks such as ImageNet-1k [4], or whether oscillatory dynamics can provide effective inductive biases for difficult combinatorial reasoning tasks such as maze pathfinding [21, 44]. This raises a fundamental question: *Can synchronization-driven dynamics serve as a scalable and parameter-efficient alternative to conventional feed-forward or transformer-style neural architectures?*

In this work, we introduce the *Winfree Oscillatory Neural Network* (*WONN*), a neural architecture built upon generalized Winfree synchronization dynamics [46, 25]. As illustrated in Fig. 1, *WONN* evolves neural representations on a high-dimensional toroidal phase space $(S^1)^d$, where each neuron is represented as a phase oscillator. Unlike Kuramoto-type models based primarily on pairwise phase differences, Winfree dynamics decompose interactions into separable sensitivity and influence functions, enabling more flexible synchronization behaviors and interaction structures.

WONN combines geometric inductive biases with flexible synchronization dynamics. The interaction functions can either be instantiated as fixed trigonometric mappings or parameterized by learnable neural networks, while grouped dynamics introduce hierarchical interactions that support both local coordination and global information flow. Under trigonometric interactions, the oscillatory core further admits an interaction energy that provides a useful stability perspective and diagnostic signal for reasoning tasks. Moreover, the dual phase–frequency state design separates fast synchronization dynamics from slower frequency evolution, enabling stable iterative refinement of representations.

We evaluate *WONN* on both visual perception and structured reasoning tasks. Across CIFAR and ImageNet benchmarks, *WONN* achieves competitive or superior accuracy compared with convolutional, transformer-based, and synchronization-inspired baselines while using substantially fewer parameters. **Importantly, *WONN* is, to our knowledge, the first synchronization-based oscillatory architecture to scale to ImageNet-1K with competitive performance against modern architectures such as ResNet and ViT.** The learned representations further exhibit characteristic bimodal phase organization, where distinct synchronized phase groups specialize to complementary coarse and fine image structures. On complex reasoning tasks, ***WONN* achieves competitive Maze-hard performance using only 1% of the parameters of the state-of-the-art *HRM* [44],** while also reaching perfect test accuracy on Sudoku with strong parameter efficiency. Taken together, these results suggest that generalized Winfree synchronization dynamics provide a scalable, high-performance, and parameter-efficient computational principle for both visual perception and logical reasoning.

2 Winfree Oscillatory Neural Networks

This section develops *WONN* by connecting classical Winfree dynamics with learnable neural architectures. We first review the underlying Winfree model and its generalizations (Sec. 2.1), then introduce the *WONN* architecture with structured interactions (Sec. 2.2), and finally discuss its key dynamical and representational properties (Sec. 2.3).

2.1 From Winfree Dynamics to Learnable Oscillatory Systems

Winfree Dynamics. We consider a population of coupled phase oscillators that are governed by the Winfree model [46]:

$$\dot{\theta}_i = \omega_i + \frac{\kappa}{N} S(\theta_i) \sum_{j=1}^N I(\theta_j), \quad i = 1, \dots, N, \quad (1)$$

where $\theta_i \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z})$ is the phase of the i -th oscillator, $\omega_i \in \mathbb{R}$ is its natural frequency and $\kappa \in \mathbb{R}_+$ is the coupling strength. The interaction is mediated through a mean-field aggregation of the influence function $I(\cdot)$, modulated by the sensitivity function $S(\cdot)$.

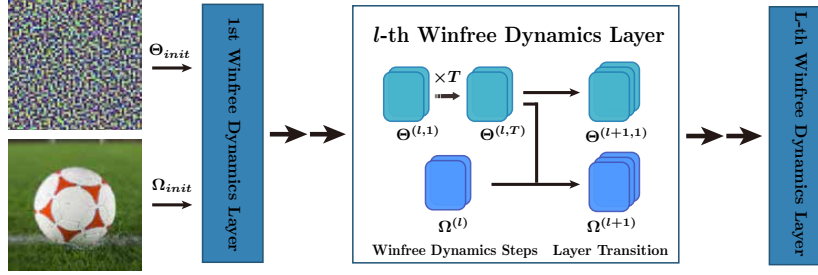


Figure 2: Overview of the *WONN* architecture. The input is encoded into an initial frequency state Ω_{init} , while the phase state Θ_{init} is initialized randomly. Computation proceeds through stacked Winfree dynamics layers, each consisting of multiple parameter-shared recurrent dynamics steps followed by layer-transition updates of both phase and frequency states. Through this iterative synchronization process, *WONN* evolves structured oscillatory representations across layers on the toroidal phase space before producing the final prediction.

Structured Interactions Beyond Kuramoto. A widely studied special case of oscillator synchronization is the Kuramoto model, whose dynamics are governed by pairwise phase differences:

$$\dot{\theta}_i = \omega_i + \gamma \sum_{j=1}^N K_{ij} \sin(\theta_j - \theta_i), \quad t > 0, \quad (2)$$

where $\gamma \in \mathbb{R}_+$ denotes the coupling strength. These dynamics preserve global phase-shift symmetry under $\theta_i \mapsto \theta_i + \alpha$, since the interaction depends only on relative phase differences. While this symmetry simplifies analysis, it may also limit the diversity of attainable collective dynamics.

By contrast, the Winfree formulation allows more general interaction structures and naturally extends beyond this symmetric regime. In particular, introducing additional coupling terms yields

$$\dot{\theta}_i = \omega_i + \gamma \sum_{j=1}^N K_{ij} [\sin(\theta_j - \theta_i) + q \sin(\theta_j + \theta_i)], \quad (3)$$

where q controls the degree of symmetry breaking. This formulation recovers the Kuramoto model when $q = 0$, and produces asymmetric interactions for $q \neq 0$, enabling richer synchronization patterns. When $q = 1$, the dynamics can be rewritten as $\dot{\theta}_i = \omega_i + 2\gamma \cos \theta_i \sum_{j=1}^N K_{ij} \sin(\theta_j)$, which corresponds to a separable Winfree form with $S(\theta) = \cos(\theta)$ and $I(\theta) = \sin(\theta)$, arising from the first-order Fourier components of the interaction functions [25].

These formulations highlight a key perspective: *collective dynamics are determined by the structure of interactions rather than pairwise phase differences alone*. This flexibility makes Winfree-type systems a natural candidate for learnable neural architectures.

Geometric structure. Each oscillator evolves on the circle $\mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z}) \cong S^1$, and the full system resides on the torus

$$\Theta = (\theta_1, \dots, \theta_d) \in \mathcal{M} := \mathbb{T}^d \cong (S^1)^d. \quad (4)$$

The dynamics define a vector field on $\mathcal{M}$, while updates can be locally represented in the tangent space $T_{\Theta}\mathcal{M} \cong \mathbb{R}^d$. In practice, this suggests performing computations in a Euclidean embedding while enforcing periodic structure in the phase variables.

2.2 Winfree Oscillatory Neural Network

Fig. 2 illustrates the overall architecture of *WONN*, where representations are computed through iterative synchronization dynamics between coupled oscillatory neurons.

State and initialization. Each layer maintains a dual state consisting of a phase variable $\Theta \in (S^1)^d$ and a frequency variable $\Omega \in \mathbb{R}^d$. The phase captures oscillatory structure, while the frequency carries input-dependent information. Given an input x , we initialize $\Omega^{(0)} = \Omega_{init} = f_{init}(x)$ and $\Theta^{(0)} = \Theta_{init} \sim \mathcal{N}(0, \sigma^2)$, where f_{init} is a learnable embedding.

Discrete Winfree dynamics. At the core of *WONN* is a discretized Winfree evolution. For each layer l , we perform T recurrent updates:

$$\theta_i^{(l,t+1)} = \theta_i^{(l,t)} + \gamma \left[\omega_i^{(l)} + S(\theta_i^{(l,t)}) \sum_j c_{ij} I(\theta_j^{(l,t)}) \right], \quad t = 1, \dots, T. \quad (5)$$

with parameters shared across time steps. Here c_{ij} denotes the coupling coefficient between oscillators i and j . The above ODE defines a controlled dynamical system over phases.

Interaction mechanisms. The behavior of the system is governed by the parameterization of the sensitivity and influence functions S and I . In the simplest case, these are defined point-wise as $S(\theta_i)$ and $I(\theta_i)$, implemented either using fixed trigonometric functions or learnable neural mappings.

To capture structured interactions, we introduce a **grouped formulation**. As shown in Fig. 3, oscillators are partitioned into spatial patches $\mathcal{G}_{p,q}$, where each patch aggregates local states into a shared influence signal:

$$I_{p,q}^{\text{patch}} = I_{\text{patch}}(\{\theta_{i,j}\}_{(i,j) \in \mathcal{G}_{p,q}}). \quad (6)$$

Here $\mathcal{G}_{p,q}$ denotes an $N \times N$ region, where N is the group size. This induces a hierarchical interaction pattern, combining intra-group synchronization with inter-group coordination.

More generally, the coupling coefficients $\sum c_{ij}$ can be implemented through local or global components. Local interactions are implemented via convolutional operators with bounded receptive fields, while global interactions are realized through attention mechanisms. This results in a heterogeneous, state-dependent mean field, extending the homogeneous coupling in classical Winfree dynamics. We use global attentive coupling for all *WONN* models throughout the paper, and discuss local convolutional coupling as an alternative in Appendix C.2.

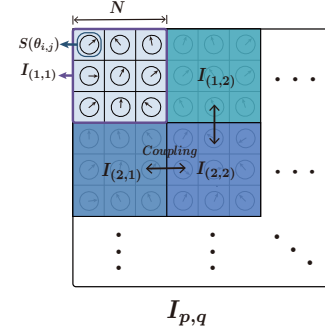


Figure 3: Illustration of S and I .

Algorithm 1 Forward pass of *WONN*

```

1: Input: input  $x$ , layers  $L$ , steps  $\{T_l\}_{l=1}^L$ , step size  $\gamma$ , group size  $N$ 
2: Output: model output  $y$ 
3:  $\Omega^{(1)} \leftarrow f_{\text{init}}(x)$ ,  $\Theta^{(1,1)} \sim \mathcal{N}(0, \sigma^2 I)$ 
4: for  $l = 1, \dots, L$  do
5:   for  $t = 1, \dots, T_l$  do
6:     (1) Point-wise Sensitivity Function
7:      $S^{(l,t)} \leftarrow S_l(\Theta^{(l,t)})$ 
8:     (2) Patch-wise Influence Function
9:     partition  $\Theta^{(l,t)}$  into non-overlapping  $N \times N$  groups  $\{\mathcal{G}_{p,q}\}$ 
10:     $I_{p,q}^{(l,t)} \leftarrow I_{\text{patch}}^l(\{\theta_{i,j}^{(l,t)}\}_{(i,j) \in \mathcal{G}_{p,q}})$ 
11:    (3) Winfree Dynamics Step
12:     $\Delta\Theta^{(l,t)} \leftarrow \Omega^{(l)} + S^{(l,t)} \odot (\sum_j c_{ij} I_{p,q}^{(l,t)})$ ,  $\Theta^{(l,t+1)} \leftarrow \text{wrap}_{[-\pi, \pi]}(\Theta^{(l,t)} + \gamma \Delta\Theta^{(l,t)})$ 
13:   end for
14:   (4) Layer Transition
15:    $\Theta^{(l+1,1)} \leftarrow \text{ThetaUpdate}_l(\Theta^{(l,T_l)})$ ,  $\Omega^{(l+1)} \leftarrow \text{OmegaUpdate}_l(\Omega^{(l)}, \Theta^{(l,T_l)})$ 
16: end for
17: (5) Output
18:  $y \leftarrow \text{OutputHead}(\Theta^{(L,T_L)})$ 
19: return  $y$ 

```

Layer transitions. After T steps, the phase and frequency are jointly updated as

$$\Theta^{(l+1)} = \text{ThetaUpdate}(\Theta^{(l,T)}), \quad \Omega^{(l+1)} = \text{OmegaUpdate}(\Omega^{(l)}, \Theta^{(l,T)}). \quad (7)$$

This design separates fast phase evolution within each layer from slower updates of the frequency state across layers, allowing oscillatory patterns to be progressively refined.

Since phases lie on the circle, computations are performed via Euclidean embeddings while preserving periodic structure. Each phase is represented as $(\sin \theta, \cos \theta)$, and the phase update `Theta-Update`

first applies weight-tied convolutional operators in this embedding space (which can be interpreted as linear operators acting on complex representation $e^{i\Theta}$) and maps the result back to the torus via

$$\Theta_{\text{new}} = \text{atan2}(\text{Conv}(\sin \Theta), \text{Conv}(\cos \Theta)) \quad (8)$$

For frequency updates, we similarly construct a tangent-space representation of Θ and combine it with the current frequency state to produce $\Omega^{(l+1)}$ through a learnable mapping. The final model representation is obtained from the phase variables $\Theta^{(L)}$ using a task-specific neural network head. Alg. 1 summarizes a single forward pass of our *WONN*.

2.3 Dynamical and Structural Properties of *WONN*

WONN exhibits several distinctive properties that emerge from its oscillatory dynamics and structured interaction design. We highlight the key properties below.

Oscillatory inductive bias. *WONN* imposes a geometric inductive bias by evolving representations on the torus $(S^1)^d$, enforcing periodic structure in the representation space. This encourages phase-based organization of features, where information can be encoded through relative phase configurations. Empirically, we observe that oscillators tend to form structured phase distributions, often exhibiting multi-modal patterns (e.g., phase bifurcation with modes separated by π). Such phase organization may correlate with meaningful feature extraction. For instance in vision tasks, we observe that different phase modes align with distinct components such as object edges (see Sec. 3.2).

Hierarchical interaction structure. The grouped Winfree formulation introduces a two-level interaction hierarchy. Within each group, oscillators synchronize through shared local aggregation, while interactions across groups enable global coordination. *This structure interpolates between purely local and fully global coupling, controlled by the group size N .* In the limiting case $N=1$, the model reduces to point-wise interactions, whereas larger groups introduce shared contextual influence. This hierarchical design allows *WONN* to adapt its interaction scale to different tasks, balancing locality and global coherence.

Flexible interaction parameterization. *WONN* allows flexible choices of the sensitivity and influence functions S and I . In particular, these functions can be instantiated either as fixed trigonometric mappings (e.g., $S(\theta) = \cos \theta$, $I(\theta) = \sin \theta$) or as learnable neural networks (e.g., MLPs). The trigonometric parameterization provides a structured and stable interaction form aligned with classical oscillator models, while the parameterization of learnable neural networks increases expressivity. This flexibility allows to trade off inductive bias and representational capacity across different tasks.

Energy structure and stability. The phase dynamics of *WONN* inherit stability properties from the underlying Winfree system. Due to the periodic geometry of $(S^1)^d$, the natural frequency term cannot, in general, be associated with a globally defined potential (see Appendix D). However, when the natural frequency is excluded and the interaction functions take separable trigonometric forms, the dynamics admit a Lyapunov function that decreases along trajectories, leading to stable synchronization patterns and attractors in phase space. In this regime, the interaction energy can be written as $E_{\text{int}} = -\sum_i \sin \theta_i \sum_j c_{ij} \sin(\theta_j)$, capturing phase alignment under the coupling coefficients. This energy also serves as a practical diagnostic: in Maze reasoning tasks (Sec. 3.3), we use it as a proxy for solution quality and perform voting to improve performance.

Separation of timescales. *WONN* separates fast oscillatory dynamics from slower feature evolution through its dual-state design. Within each layer, the phase state Θ undergoes multiple recurrent updates, enabling rapid synchronization and local reorganization on $(S^1)^d$. In contrast, the frequency state Ω is updated once per layer, acting as a slower carrier of information across layers. This separation of timescales decouples transient oscillatory computation from cross-layer feature transformation, which is reminiscent of hierarchical processing in biological systems [13, 15, 9, 44].

3 Experiments

3.1 Experimental Setup

We evaluate *WONN* on a diverse set of benchmarks spanning visual perception and logical reasoning. The visual tasks include CIFAR-10/100 [19] and ImageNet-100/1K [4], while the reasoning tasks consist of Maze-hard pathfinding [21, 44] and Sudoku solving [45].

Table 1: Image classification results on **CIFAR-10/100**. $\dagger$ ViT models are trained with additional regularization (e.g., CutMix and label smoothing), while other models follow a plain training protocol.

Models	CIFAR-10		CIFAR-100	
	Accuracy (%)	# Parameters	Accuracy (%)	# Parameters
ResNet-18	93.48 $\pm$ 0.16	11.17M	70.53 $\pm$ 0.10	11.22M
ResNet-50	94.22 $\pm$ 0.14	23.52M	73.54 $\pm$ 0.41	23.71M
ViT-T†	90.34 $\pm$ 0.18	5.36M	67.59 $\pm$ 0.44	5.38M
ViT-S†	93.43 $\pm$ 0.29	21.34M	71.18 $\pm$ 0.62	21.38M
ViT-B†	92.04 $\pm$ 0.25	85.15M	71.05 $\pm$ 0.40	85.22M
AKOrNattn	93.66 $\pm$ 0.17	4.60M	72.03 $\pm$ 0.34	4.62M
<i>$S(\theta_i) \& I(\theta_i)$ as MLPs</i>				
WONN (Ch = 128 $\rightarrow$ 128)	94.55 $\pm$ 0.09	3.08M	73.77 $\pm$ 0.40	3.09M
WONN (Ch = 64 $\rightarrow$ 256)	95.12 $\pm$ 0.04	7.54M	75.12 $\pm$ 0.49	7.56M
WONN (Ch = 256 $\rightarrow$ 256)	95.24 $\pm$ 0.12	12.02M	76.20 $\pm$ 0.45	12.04M
<i>$S(\theta_i) \& I(\theta_i)$ as trigonometric functions</i>				
WONN (Ch = 128 $\rightarrow$ 128)	94.50 $\pm$ 0.15	2.98M	74.48 $\pm$ 0.16	3.00M
WONN (Ch = 64 $\rightarrow$ 256)	95.08 $\pm$ 0.07	7.40M	75.81 $\pm$ 0.33	7.43M
WONN (Ch = 256 $\rightarrow$ 256)	95.26 $\pm$ 0.05	11.84M	76.17 $\pm$ 0.52	11.86M

Visual Perception. For image classification, we benchmark *WONN* against representative convolutional and transformer-based architectures, including *ResNet-18* and *ResNet-50* [14], as well as Vision Transformers (*ViT-Tiny/Small/Base*) [5, 41]. We also include *AKOrN* [26], a recent synchrony-based model, as a closely related baseline. *All methods are trained under standard, architecture-appropriate protocols, and no additional training techniques are introduced to favor WONN, ensuring a fair comparison.*

Logical Reasoning. For reasoning tasks, we evaluate *WONN* on Maze-hard and Sudoku. On Maze-hard, we mainly compare against both general-purpose LLMs (e.g., *DeepSeek-R1* [11]) and specialized recurrent reasoning architectures, including *HRM* [44] and *TRM* [16]. On Sudoku, we compare with *AKOrN* [26] and several other neural reasoning baselines [45, 30, 48].

Other implementation details and extended experimental results are referred to in Appendix C.

3.2 Image Classification on CIFAR and ImageNet

We evaluate *WONN* on standard image classification benchmarks, including CIFAR-10/100 and ImageNet-100/1K. We consider two parameterizations of the interaction functions S and I : (i) learnable MLP-based mappings and (ii) fixed trigonometric mappings. All models use grouped Winfree dynamics in Eq. (6) with group size $N = 2$. To assess the scalability of model size, we study three model sizes with approximately 3M, 7.5M, and 12M parameters.

Results on CIFAR-10/100. Table 1 compares *WONN* with *ResNet*, *ViT*, and *AKOrN*, with results averaged over three random seeds. *WONN* consistently outperforms all baselines on both CIFAR-10 and CIFAR-100 while using substantially fewer parameters. Notably, *WONN*, trained with a plain protocol, surpasses transformer-based baselines that rely on additional regularization. These results demonstrate the effectiveness and parameter efficiency of oscillatory dynamics.

Results on ImageNet-100/1K. Table 2 shows that the trends observed on CIFAR persist at larger scale: *WONN* achieves competitive accuracy with substantially fewer parameters than standard baselines. In particular, *WONN* successfully scales to ImageNet-1K, surpassing or matching *ViT/ResNet* performance using roughly half or fewer parameters. *To the best of our knowledge, this is the first synchrony-based architecture that scales effectively to large-scale image classification.*

Two-peak phase distributions. To better understand the internal phase organization of *WONN*, we analyze the final θ distribution in Fig. 4 left. Interestingly, the phase variables consistently bifurcate into two dominant modes, producing a characteristic bimodal distribution with peaks approximately separated by π . To visualize the functional role of these modes, we construct weighted activation maps by measuring the proximity of each oscillator phase to the corresponding peak.

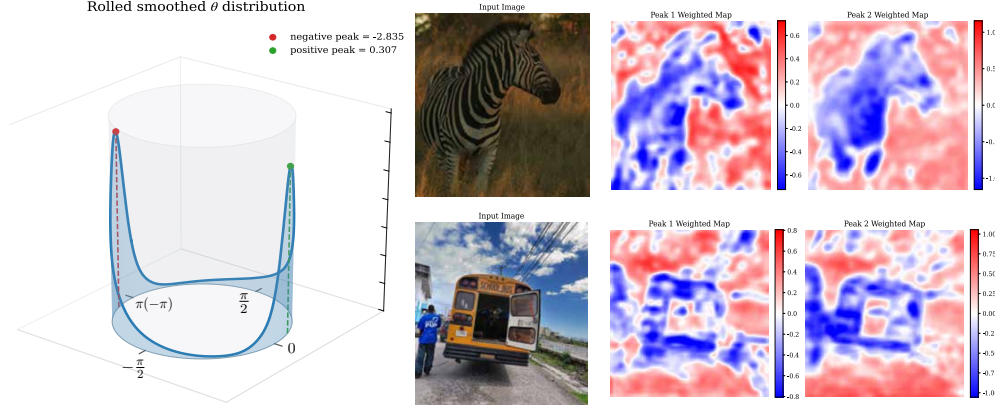


Figure 4: Visualization of the final phase distribution in *WONN* on image recognition tasks. **Left:** the phase variables θ exhibit a characteristic bimodal distribution, with two dominant peaks approximately separated by π . **Right:** heatmaps constructed by weighting oscillators according to their proximity to each phase peak. The two modes capture complementary object structures: one emphasizes coarse global regions, while the other focuses on finer local details and boundaries. Together, the synchronized phase organization produces structured object-aware representations.

Table 2: Image classification results on **ImageNet-100** and **ImageNet-1K**. $\dagger$ ViT models are trained with additional regularization techniques compared to other models.

Models	ImageNet-100		ImageNet-1k	
	Accuracy(%)	# Parameters	Accuracy(%)	# Parameters
ResNet-18	78.20	11.23M	69.73	11.69M
ResNet-50	81.18	23.71M	76.89	25.56M
ViT-S-16†	77.96	21.70M	75.54	22.05M
ViT-B-16†	76.36	85.87M	75.85	86.57M
AKOrNattn	80.08	4.62M	67.45	4.85M
<i>S(θ_i)&I(θ_i) as MLPs</i>				
WONN (Ch = 128 $\rightarrow$ 128)	81.50	3.09M	—	—
WONN (Ch = 64 $\rightarrow$ 256)	82.04	7.56M	74.84	7.79M
WONN (Ch = 256 $\rightarrow$ 256)	82.88	12.05M	76.78	12.28M
<i>S(θ_i)&I(θ_i) as trigonometric functions</i>				
WONN (Ch = 128 $\rightarrow$ 128)	81.76	3.00M	—	—
WONN (Ch = 64 $\rightarrow$ 256)	82.56	7.43M	—	—
WONN (Ch = 256 $\rightarrow$ 256)	82.22	11.87M	—	—

Fig. 4 right reveals that the two synchronized phase modes capture *complementary aspects of the object structure*. One phase mode primarily responds to coarse, global object regions, while the other emphasizes finer local structures and boundaries. Despite different spatial characteristics, both modes remain highly aligned with semantically relevant object regions. This suggests that the oscillatory synchronization dynamics naturally induce a structured phase decomposition of representations, where different synchronized phase groups specialize to distinct levels of spatial abstraction.

Notably, the emergence of such bimodal phase organization is not explicitly imposed by any architecture or supervision objective, but instead arises spontaneously from the collective synchronization dynamics. We hypothesize that this self-organized phase separation contributes to the strong recognition performance and parameter efficiency of *WONN* by enabling coordinated yet diverse feature representations across oscillators. Additional visualizations are provided in Appendix E.1.

3.3 Maze-hard Pathfinding

We evaluate the reasoning capability of *WONN* on the **Maze-hard** benchmark, which requires finding optimal paths in 30×30 mazes and tests long-horizon search and algorithmic reasoning.

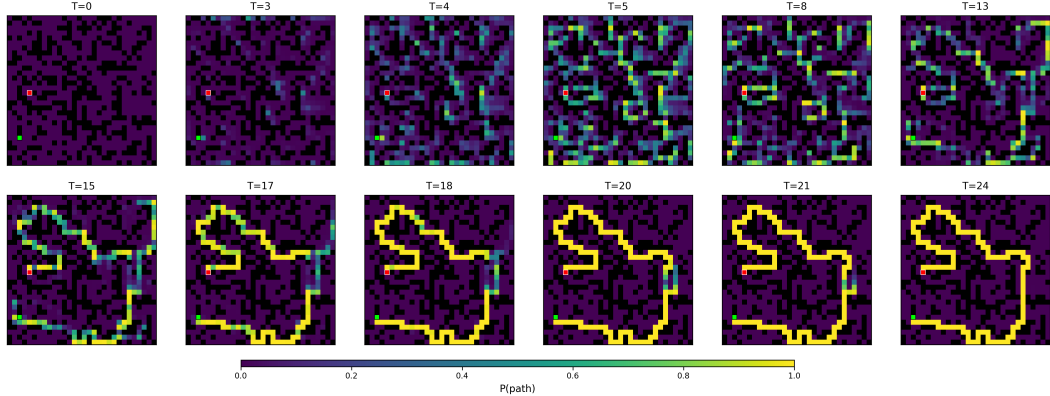


Figure 5: Synchronous path formation on Maze-hard. Each panel shows the predicted per-cell path probability. *WONN* initially produces multiple diffuse path fragments. As the dynamics evolve, these fragments progressively synchronize and merge, forming a coherent path from start to goal.

Following [44], we construct 1,000 training and 1,000 test instances whose difficulty levels exceed 110. For this task, *WONN* uses point-wise interactions ($N = 1$).

Table 3: Maze-hard pathfinding results. *Energy Voting* denotes selecting the solution with the lowest energy among 32 independently sampled trajectories.

Models	Accuracy (%)	# Parameters
<i>LLMs</i>		
Deepseek R1	0.0	671B
Claude 3.7 8K	0.0	?
O3-mini-high	0.0	?
<i>Recurrent models</i>		
HRM	74.5	27M
TRM-Att (dihedral augmented)	85.3	7M
TRM-MLP (dihedral augmented)	0.0	19M
<i>Other synchrony-based model</i>		
<i>AKOrN</i>^{attn}	36.2	1M
<i>$S(\theta_i)$ & $I(\theta_i)$ as trigonometric functions</i>		
<i>WONN</i>	76.2	0.396M
<i>WONN</i> (Energy Voting)	80.1	0.396M

Results on Maze-hard. Table 3 shows that *WONN* significantly outperforms LLM-based baselines and the synchrony-based model *AKOrN*, and surpasses the strong recurrent model *HRM*. Despite using only **0.396M** parameters (**1.47%** of *HRM*), *WONN* achieves **76.2%** accuracy. Importantly, leveraging the energy structure of the dynamics, we further improve performance via *energy voting*: by sampling 32 trajectories and selecting the solution with the lowest final energy, accuracy increases to **80.1%**. This highlights the practical usage of the energy formulation as a *test-time selection criterion*. Taken together, these results demonstrate that structured oscillatory dynamics provide an effective and highly parameter-efficient inductive bias for long-horizon reasoning.

Synchronous Pathfinding on Maze-hard. To better understand the role of oscillatory dynamics, we visualize the intermediate states of *WONN* on Maze-hard (Fig. 5). We plot heatmaps of the predicted per-cell path probability, revealing how solutions emerge over time.

Unlike conventional methods that construct paths incrementally or greedily, *WONN* exhibits a distinct synchronization-driven process. In the early stage, the model generates multiple diffuse path fragments across the maze that correspond to candidate solution trajectories. These fragments are not locally consistent and may include invalid transitions. As the dynamics evolve, these candidate fragments begin to interact and synchronize. Inconsistent paths are suppressed, while compatible segments reinforce each other and progressively merge. This collective refinement process leads

to the emergence of a single coherent path connecting the start and goal. In later stages, the model further refines the solution by removing redundant segments and improving path optimality.

Additional examples (Appendix E.2) show that this multi-fragment-to-single-path behavior is consistent across hard instances. These observations suggest that synchronization dynamics provide a powerful inductive bias for combinatorial search: *instead of committing to a single trajectory early, the model explores multiple candidates in parallel and resolves them through collective alignment.*

3.4 Sudoku Reasoning

We further evaluate *WONN* on Sudoku, a standard benchmark for combinatorial reasoning. We use the dataset from [45], where training instances contain 31–42 given digits, and evaluation is performed on 1,000 test boards drawn from the same distribution. For this task, *WONN* adopts point-wise interactions ($N = 1$) and follows the same training protocol as *AKOrN*.

Results on Sudoku. Table 4 reports results averaged over five runs. *WONN* achieves perfect accuracy on the test set, matching the best-performing methods. It improves upon *AKOrN* and *HRM* while using significantly fewer parameters, and achieves comparable performance to both recurrent and transformer-based baselines [30, 48]. Consistent with the Maze-hard results, these findings suggest that oscillatory dynamics provide an effective inductive bias for structured reasoning tasks.

Table 4: Sudoku solving results.

Models	Accuracy (%)	# Parameters
SAT-Net	98.3	0.618M
RRN	99.8	0.201M
R-Transformer	100.0	0.211M
Transformer	98.6 $\pm$ 0.3	16.87M
HRM	99.7 $\pm$ 0.2	27.28M
AKOrN^{attn} ($T = 16$)	99.8 $\pm$ 0.1	2.98M
<i>S(θ_i) & I(θ_i) as trigonometric functions</i>		
WONN ($T = 16$)	100.0$\pm$ 0.0	1.58M

4 Conclusion

We introduced the *Winfree Oscillatory Neural Network (WONN)*, a neural architecture built upon generalized Winfree synchronization dynamics. Unlike conventional architectures that primarily rely on static feature transformations, *WONN* performs computation through the collective evolution and synchronization of phase oscillators on a toroidal state space $(S^1)^d$. By combining flexible interaction parameterizations, hierarchical grouped synchronization dynamics, and a dual phase–frequency state design, *WONN* provides a scalable framework for oscillatory neural computation.

Empirically, we show that synchronization-based architectures can scale far beyond the limited settings previously explored in oscillatory or synchronization-inspired models. On visual perception tasks, *WONN* achieves competitive or superior performance on CIFAR and ImageNet benchmarks while maintaining strong parameter efficiency. In particular, *WONN* successfully scales to ImageNet-1K, which, to the best of our knowledge, makes it the first synchronization-based oscillatory architecture to achieve competitive large-scale image recognition performance. The learned phase representations further exhibit characteristic bimodal phase organization, where distinct synchronized phase groups specialize to complementary coarse and fine image structures. Beyond perception, *WONN* also demonstrates strong capability on structured reasoning tasks. On Maze-hard, *WONN* achieves competitive performance using only a tiny fraction of the parameters of state-of-the-art recurrent reasoning models. Visualization of the temporal evolution further reveals that the network progressively assembles local candidate path fragments into coherent global solutions, suggesting that synchronization provides an effective inductive bias for complex search and reasoning problems. On Sudoku, *WONN* further achieves perfect test accuracy with strong parameter efficiency.

More broadly, our results suggest that synchronization dynamics may provide a promising alternative computational paradigm for modern machine learning. We hope this work encourages further exploration of oscillatory neural computation, synchronization-driven representation learning, and dynamical systems perspectives on large-scale neural architectures.

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A Related Work

Oscillatory neural computation. Oscillatory activity is widely regarded as a fundamental organizing principle in biological neural systems, supporting temporal coordination, communication, and structured information processing [2, 8, 27]. This perspective has motivated the use of oscillatory dynamics as inductive biases in artificial neural networks, where oscillations provide a temporal scaffold for computation. Early work introduced rhythmic gating mechanisms for sparse and event-based processing [28], while more recent approaches have explored dynamical systems-based architectures, including stable recurrent networks and graph-based models, where oscillatory or wave-like dynamics enable long-range interactions and memory [33, 32, 18, 36, 35, 17]. More broadly, oscillatory mechanisms have been shown to improve learning dynamics, robustness, and representational organization by structuring computation through phase and temporal evolution [6, 7, 38].

Synchronization-inspired models in machine learning. While oscillations structure computation in time, synchronization governs how multiple oscillators interact through collective phase alignment. Building on this principle, a growing body of work has incorporated synchronization into neural architectures. Prior approaches have leveraged phase or complex-valued representations to organize object-centric features and latent structure [24, 39, 23, 10], introduced synchronization mechanisms into modern architectures through bounded-confidence interactions [22], and explored neural synchronization as a latent representation in recurrent dynamical architectures such as the Continuous Thought Machine [3]. Synchrony-based signals have also been used as training-time plasticity mechanisms in spiking neural networks, for example through dopamine-modulated spike-synchrony-dependent plasticity [40]. Closely related to our work are oscillator-based neural systems such as *AKOrN* [26]. At a more fundamental level, these approaches are closely related to classical coupled-oscillator systems, particularly the Kuramoto and Winfree models [20, 46]. Kuramoto-type dynamics based on pairwise phase interactions have been widely explored in cortical oscillation modeling, learnable oscillator systems, graph neural networks, orientation diffusion models, and phase-based neural architectures [1, 31, 29, 37, 47]. In contrast, Winfree-type formulations introduce separable sensitivity–influence interactions, enabling more flexible and expressive collective dynamics [12, 25]. *WONN* builds on this formulation by using generalized Winfree dynamics with flexible parameterizations and grouped interactions, enabling hierarchical information flow through synchronization dynamics on a toroidal phase space.

B Discussions

B.1 Limitation and Future Work

Our results suggest that oscillatory synchronization can serve not only as a biological or physical metaphor, but also as a practical computational principle for modern neural networks. At the same time, *WONN* represents only an initial step toward synchronization-driven large-scale neural computation. While the recurrent Winfree dynamics provide strong parameter efficiency and expressive iterative refinement, they also introduce additional computational cost compared with conventional feed-forward architectures such as ResNets and ViTs, since each layer involves multiple recurrent synchronization steps. Furthermore, although *WONN* demonstrates competitive scalability up to ImageNet-1K, exploring larger-scale models and even language models remains an important direction for future work. Future research may further investigate more efficient synchronization mechanisms, adaptive or continuous-time dynamics, larger-scale oscillatory architectures, and deeper theoretical understanding of the stability, geometry, and representation structure induced by Winfree-type neural dynamics.

B.2 Difference against *AKOrN*

WONN is closely related to *AKOrN* [26] in that both methods introduce oscillatory dynamics into neural architectures. However, the two models differ in several essential aspects.

Practical scalability. In Table 7, we additionally compare the practical training cost of *WONN* and *AKOrN* on ImageNet-1K. Despite using a batch size twice as large and 8 powerful H200 GPUs, *AKOrN* requires nearly 1,000 GPU hours to train its standard-size model for only 200 epochs, whereas *WONN* is trained for 300 epochs using only 4 H100 GPUs with substantially lower total GPU hours. We also observe a large memory gap under the same per-GPU batch size: *AKOrN* requires more

than 110GB aggregate GPU memory, compared with approximately 35GB for *WONN*. These results suggest that *WONN* is considerably more favorable for large-scale training under realistic compute and memory constraints.

Different state geometries. *AKOrN* represents each oscillatory neuron by a high-dimensional unit vector, so its oscillator state is embedded in a Euclidean ambient space. In contrast, *WONN* represents the oscillator state directly by a phase angle. Thus, each scalar oscillator evolves on the circle S^1 , and a collection of d oscillators naturally lives on the torus $(S^1)^d$. This makes the periodic geometry of the representation explicit rather than treating oscillation through a vector-valued Euclidean embedding.

Separable interaction mechanisms. *AKOrN* is motivated by Kuramoto-type synchronization (Eq. 2), where interactions are governed primarily by pairwise phase differences. Such dynamics are invariant under global phase shifts, since replacing every phase by $\theta_i + \alpha$ leaves the differences $\theta_j - \theta_i$ unchanged. While this symmetry simplifies the interaction structure, it may also restrict the class of admissible dynamics. In contrast, *WONN* builds upon the more general Winfree formulation (Eq. 1), where oscillator updates are decomposed into a sensitivity function of the receiving oscillator and an influence function of neighboring oscillators. This formulation recovers Kuramoto-style phase-difference coupling as a special case, while also permitting symmetry-breaking interactions (Eq. 3) that depend on absolute phase. As a result, *WONN* defines a more flexible and expressive class of oscillatory interactions.

Grouped hierarchical interactions. *WONN* extends the classical Winfree formulation into a grouped neural architecture. Instead of applying only pointwise oscillator interactions, *WONN* partitions oscillators into groups and computes group-level influence signals. This grouped formulation induces a hierarchical interaction structure: oscillators may synchronize locally within a group, while group-level influence signals coordinate information across larger spatial or feature regions. The group size therefore controls the interaction scale, interpolating between pointwise dynamics and more structured collective dynamics.

Flexible interaction parameterization. *WONN* provides a flexible parameterization of the Winfree interaction functions. The sensitivity and influence functions can either be instantiated as fixed trigonometric mappings, such as $S(\theta) = \cos \theta$ and $I(\theta) = \sin \theta$, or implemented as learnable neural networks. The former preserves a strong oscillator-inspired inductive bias, while the latter increases representational flexibility and expressiveness. This allows the interaction dynamics of *WONN* to adapt naturally to different tasks, datasets, and computational regimes.

C Additional Implementation Details and Extended Experiment Results

C.1 Implementation Details

C.1.1 IMAGE RECOGNITION

For Image Recognition, we train all the models under comparable training protocols. The detailed training recipes are reported in Table 5 for **CIFAR-10/100**, Table 6 for **ImageNet-100**, and Table 7 for **ImageNet-1K**.

Table 5: Implementation details on **CIFAR-10/100**.

Hyperparameters	Models				
	<i>WONN</i>	<i>AKOrN</i> ^{attn}	ResNet-18/50	ViT-T/S	ViT-B
Batch Size	64	128	128	256	256
Optimizer	AdamW	Adam	AdamW	AdamW	AdamW
Weight Decay	0.0001	–	0.0001	0.05	0.05
LR	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$1 \cdot 10^{-4}$
LR Decay	$\times$	$\times$	$\times$	cosine	cosine
Min LR	$\times$	$\times$	$\times$	0	0
Warmup Epochs	$\times$	$\times$	$\times$	10	10
Epochs	200	200	200	300	300
Dynamics Layers (L)	6	3	–	–	–
Dynamics Time Step (T)	3	3	–	–	–
Group Size (N)	2	–	–	–	–
Oscillator Dimension (n)	–	4	–	–	–
GPUs	1×H100	1×H100	1×H100	1×H100	1×H100
Dropout	$\times$	$\times$	$\times$	$\times$	$\times$
Label Smoothing	$\times$	$\times$	$\times$	0.1	0.1
Mixup Alpha	$\times$	$\times$	$\times$	0.8	0.8
Cutmix Alpha	$\times$	$\times$	$\times$	1.0	1.0
Mixup Prob.	$\times$	$\times$	$\times$	0.2	0.2
Cutmix Prob.	$\times$	$\times$	$\times$	0.8	0.8

Table 6: Implementation details on **ImageNet-100**.

Hyperparameters	Models					
	<i>WONN</i>	<i>AKOrN</i> ^{attn}	ResNet-18	ResNet-50	ViT-S-16	ViT-B-16
Batch Size	64	16	128	128	256	256
Optimizer	AdamW	Adam	AdamW	AdamW	AdamW	AdamW
Weight Decay	0.0001	–	0.0001	0.0001	0.05	0.05
LR	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$2/3 \cdot 10^{-4}$
LR Decay	$\times$	$\times$	$\times$	$\times$	cosine	cosine
Min LR	$\times$	$\times$	$\times$	$\times$	0	0
Warmup Epochs	$\times$	$\times$	$\times$	$\times$	10	10
Epochs	200	200	200	200	300	300
Dynamics Layers (<i>L</i>)	6	3	–	–	–	–
Dynamics Time Step (<i>T</i>)	3	3	–	–	–	–
Group Size (<i>N</i>)	2	–	–	–	–	–
Oscillator Dimension (<i>n</i>)	–	4	–	–	–	–
Input Patch Size	7	7	–	–	16	16
GPUs	1×H100	1×H100	1×H100	1×H100	1×H100	1×H100
Dropout	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
Label Smoothing	$\times$	$\times$	$\times$	$\times$	0.1	0.1
Mixup Alpha	$\times$	$\times$	$\times$	$\times$	0.8	0.8
Cutmix Alpha	$\times$	$\times$	$\times$	$\times$	1.0	1.0
Mixup Prob.	$\times$	$\times$	$\times$	$\times$	0.2	0.2
Cutmix Prob.	$\times$	$\times$	$\times$	$\times$	0.8	0.8

Table 7: Implementation details on **ImageNet-1K**.

Hyperparameters	Models					
	<i>WONN</i>	<i>AKOrN</i> ^{attn}	ResNet-18	ResNet-50	ViT-S-16	ViT-B-16
Batch Size	512	512	128	128	1024	1024
Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW	AdamW
Weight Decay	0.001	0.001	0.001	0.001	0.05	0.05
LR	$7.5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	$2/3 \cdot 10^{-4}$
LR Decay	cosine	cosine	cosine	cosine	cosine	cosine
Min LR	$1 \cdot 10^{-6}$	$1 \cdot 10^{-6}$	$1 \cdot 10^{-6}$	$1 \cdot 10^{-6}$	0	0
Warmup Epochs	10	10	10	10	10	10
Epochs	300	200	200	200	300	300
Dynamics Layers (L)	6	3	—	—	—	—
Dynamics Time Step (T)	3	3	—	—	—	—
Group Size (N)	2	—	—	—	—	—
Oscillator Dimension (n)	—	4	—	—	—	—
Input Patch Size	7	7	—	—	16	16
GPUs	4×H100	8×H200	1×H100	1×H100	4×H100	4×H100
GPU hours	750	950	—	—	—	—
Dropout	X	X	X	X	X	X
Mixup Alpha	—	—	—	—	0.8	0.8
Cutmix Alpha	1.0	1.0	1.0	1.0	1.0	1.0
Label Smoothing	0.1	0.1	0.1	0.1	0.1	0.1
Mixup Prob.	0.0	0.0	0.0	0.0	0.2	0.2
Cutmix Prob.	0.8	0.8	0.8	0.8	0.8	0.8

C.1.2 SOLVING MAZE-HARD

For the Maze-hard task, we train the two models *WONN* and *AKOrN* under the same training protocol, with implementation details summarized in Table 8. Results for the remaining baselines, including large language models and recurrent models, are taken from [44, 16].

Table 8: Implementation details on **Maze-hard**.

Hyperparameters	Models	
	<i>WONN</i>	<i>AKOrN</i> ^{attn}
Batch Size	64	64
Channel	128	256
Optimizer	Adam	Adam
Weight Decay	—	—
LR	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
LR Decay	—	—
Min LR	—	—
Warmup Epochs	—	—
Epochs	9000	9000
Dynamics Layers (L)	1	1
Dynamics Time Step (T)	24	24
Group Size (N)	1	—
GPUs	1×H100	1×H100

C.1.3 SOLVING SUDOKU

For the Sudoku task, we train *WONN* following the same training protocol as *AKOrN*. The detailed configuration is summarized in Table 9. Results for the remaining baselines are taken from prior work [45, 30, 26].

Table 9: Implementation details on **Sudoku**.

Hyperparameters	Models		
	<i>WONN</i>	<i>AKOrN</i> ^{attn}	<i>HRM</i>
Batch Size	100	100	384
Channel	256	512	512
Optimizer	Adam	Adam	Adam-atan2
Weight Decay	—	—	1.0
LR	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-4}$
LR Decay	—	—	—
Min LR	—	—	—
Warmup Epochs	—	—	2000 steps
Epochs	100	100	2000
Dynamics Layers (L)	1	1	$H/L = 4/4$
Dynamics Time Step (T)	16	16	—
Group Size (N)	1	—	—
GPUs	1×H100	1×H100	1×H100

C.2 Additional Results and Ablations on Image Recognition

Here we provide additional experimental results for image recognition benchmarks, including supplementary evaluations on ImageNet-100 (Table 10) and CIFAR-100 (Table 11). We further conduct several ablation studies on CIFAR-100 to analyze the contribution of key architectural and dynamical components of *WONN*. Throughout this section, L denotes the number of Winfree dynamics layers, T denotes the number of recurrent dynamics steps per layer, and N denotes the group size. Tables 10 and 11 summarize the ablation results on ImageNet-100 and CIFAR-100, respectively.

Table 10: Additional image classification results on **ImageNet-100**.

Models	Accuracy (%)	# Parameters
<i>AKOrN</i> ^{attn} (input patch size = 7)	75.12	4.63M
<i>AKOrN</i> ^{attn} (input patch size = 4)	80.08	4.62M
<i>S</i> (θ_i) and <i>I</i> (θ_i) as MLPs, input patch size = 8		
<i>WONN</i> (Ch = 256 → 256)	82.08	12.09M
<i>S</i> (θ_i) and <i>I</i> (θ_i) as MLPs, input patch size = 7		
<i>WONN</i> (Ch = 128 → 128)	78.78	3.11M
<i>WONN</i> (Ch = 64 → 256)	79.12	7.57M
<i>WONN</i> (Ch = 256 → 256)	80.08	12.08M
<i>S</i> (θ_i) and <i>I</i> (θ_i) as MLPs, input patch size = 4		
<i>WONN</i> (Ch = 128 → 128)	81.50	3.09M
<i>WONN</i> (Ch = 64 → 256)	82.04	7.56M
<i>WONN</i> (Ch = 256 → 256)	82.88	12.05M
<i>S</i> (θ_i) and <i>I</i> (θ_i) as trigonometric functions, input patch size = 7		
<i>WONN</i> (Ch = 128 → 128)	77.92	3.01M
<i>WONN</i> (Ch = 64 → 256)	78.78	7.44M
<i>WONN</i> (Ch = 256 → 256)	80.34	11.89M
<i>S</i> (θ_i) and <i>I</i> (θ_i) as trigonometric functions, input patch size = 4		
<i>WONN</i> (Ch = 128 → 128)	81.76	3.00M
<i>WONN</i> (Ch = 64 → 256)	82.56	7.43M
<i>WONN</i> (Ch = 256 → 256)	82.22	11.87M

Table 11: Additional image classification results on **CIFAR-100**.

Models	Accuracy (%)	# Parameters
<i>$S(\theta_i)$ & $I(\theta_i)$ as MLPs, $L = 6$, $T = 3$, $N = 2$</i>		
WONN (Ch = 128 $\rightarrow$ 128)	73.77 $\pm$ 0.40	3.09M
WONN (Ch = 64 $\rightarrow$ 256)	75.12 $\pm$ 0.49	7.56M
WONN (Ch = 256 $\rightarrow$ 256)	76.20 $\pm$ 0.45	12.04M
<i>$S(\theta_i)$ & $I(\theta_i)$ as trigonometric functions, $L = 6$, $T = 3$, $N = 2$</i>		
WONN (Ch = 128 $\rightarrow$ 128)	74.48 $\pm$ 0.16	3.00M
WONN (Ch = 64 $\rightarrow$ 256)	75.81 $\pm$ 0.33	7.43M
WONN (Ch = 256 $\rightarrow$ 256)	76.17 $\pm$ 0.52	11.86M
<i>$S(\theta_i)$ & $I(\theta_i)$ as MLPs, Ch=256, local (conv) and globle (attn) coupling</i>		
WONN ^{attn} (L1T3N2)	66.14	1.65M
WONN ^{attn} (L6T1N2)	74.53	12.04M
WONN ^{attn} (L6T3N1)	75.93	11.82M
WONN ^{attn} (L6T3N2)	76.20	12.04M
WONN ^{conv} (L6T3N2)	75.63	18.72M
WONN ^{attn} (L6T3N4)	75.13	15.14M
WONN ^{attn} (L6T6N2)	76.40	12.04M
WONN ^{attn} (L9T3N2)	76.77	18.28M
<i>$S(\theta_i)$ & $I(\theta_i)$ as trigonometric functions, Ch=256, local (conv) and globle (attn) coupling</i>		
WONN ^{attn} (L1T3N2)	66.07	1.62M
WONN ^{attn} (L6T1N2)	74.58	11.86M
WONN ^{attn} (L6T3N2)	76.17	11.86M
WONN ^{attn} (L6T3N1)	76.94	11.79M
WONN ^{conv} (L6T3N2)	76.27	18.54M
WONN ^{attn} (L6T3N4)	76.24	12.67M

Impact of Patch Size. Table 10 studies the input patch size on ImageNet-100. For both MLPs-based and trigonometric interaction functions, a smaller patch size generally improves performance, suggesting that finer spatial tokenization might provides more effective oscillatory representations.

Impact of Interaction Parameterization. We compare two choices for the sensitivity and influence functions, namely learnable MLPs and fixed trigonometric functions. Both variants achieve strong performance, indicating that the benefit of **WONN** mainly comes from the oscillatory dynamical structure rather than a specific parameterization. The trigonometric form provides a stronger oscillator-inspired inductive bias and tends to outperform in small datasets, while the learnable MLP parameterization offers greater flexibility, which may be beneficial for scaling to larger and more complex datasets.

Impact of Depth (L). Increasing the number of Winfree dynamics layers generally improves performance. For example, moving from shallow layers to $L = 6$ substantially improves CIFAR-100 accuracy, showing that stacked dynamical refinement is important for representation learning.

Impact of Dynamics Steps (T). Increasing the number of recurrent dynamics steps also tends to improve performance, but at the cost of additional computation. This suggests a trade-off between the evolution horizon of synchronization dynamics and computational efficiency.

Impact of Group Size (N). The effect of group size depends on the interaction form. MLP-based interactions perform best at $N = 2$, while trigonometric interactions benefit from a different grouping choice, with $N = 1$ outperforming $N = 2$ among grouped variants. This suggests that selecting an appropriate group size can enhance **WONN** performance by matching the interaction scale to the parameterization of the interaction function.

Impact of Coupling Type (conv vs. attn). We compare global attentive coupling with local convolutional coupling under both MLP-based and trigonometric interactions. Although convolutional coupling achieves comparable accuracy, it introduces substantially more parameters in our implementation. Attentive coupling therefore offers a more efficient mechanism for coordinating oscillators, and its better parameter efficiency may make it more favorable for scaling **WONN** to larger and more complex datasets.

C.3 Solving Maze-hard

Empirically, we observe that the interaction energy E_{int} consistently decreases as the number of time steps T increases, serving as a reliable metric for the degree of synchronization (as detailed in Sec. 2.3). Leveraging this property, we introduce the energy-based voting mechanism during inference to enhance model performance. Specifically, we appropriately extend the evaluation time steps T and perform K independent trials with different random initializations of initial phases. The prediction corresponding to the lowest final interaction energy E_{int} is ultimately selected as the optimal solution, which empirically boosts the performance.

Table 12: Solving Maze-hard results

Models	Accuracy (%)	# Parameters
<i>Other synchrony-based model</i>		
AKOrN ^{attn} (L1T16)	32.3	1M
AKOrN ^{attn} (L1T24)	36.2	1M
<i>S(θ_i) & I(θ_i) as trigonometric functions, L=1, T=24</i>		
WONN	76.2	0.396M
WONN ($T_{eval} = 25, K=32$)	80.1	0.396M

D Energy Structure and Topological Obstruction

We clarify the energy structure of the trigonometric Winfree dynamics used in **WONN**. For simplicity, we analyze the continuous-time phase dynamics

$$\dot{\theta}_i = \omega_i + \gamma \cos \theta_i \sum_j c_{ij} \sin \theta_j, \quad \theta_i \in \mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z}), \quad (9)$$

where ω_i is the natural frequency, $\gamma > 0$ is the interaction strength, and c_{ij} is the coupling coefficient from oscillator j to oscillator i .

Absence of a global potential in the presence of natural frequencies. We first show that the natural-frequency term prevents the full dynamics from being written as a globally defined gradient flow on the torus. Consider the pure frequency dynamics

$$\dot{\theta}_i = \omega_i. \quad (10)$$

If this vector field were the negative gradient of a globally defined scalar potential $E : \mathbb{T}^d \rightarrow \mathbb{R}$, then in local phase coordinates it would satisfy

$$-\frac{\partial E}{\partial \theta_i} = \omega_i, \quad \text{or equivalently} \quad \frac{\partial E}{\partial \theta_i} = -\omega_i. \quad (11)$$

Locally, this implies

$$E(\Theta) = -\sum_i \omega_i \theta_i + \text{const}. \quad (12)$$

However, θ_i is not a globally single-valued function on $\mathbb{T}$. Since θ_i and $\theta_i + 2\pi$ represent the same point on the circle, a globally defined potential must be invariant under the identification

$$\theta_i \sim \theta_i + 2\pi.$$

But the local potential changes by

$$E(\Theta + 2\pi e_i) - E(\Theta) = -2\pi\omega_i. \quad (13)$$

Therefore, when $\omega_i \neq 0$, this local potential is not single-valued on the torus. Equivalently, the constant frequency drift has nonzero circulation along the fundamental cycle of the i -th circle:

$$\oint \omega_i d\theta_i = 2\pi\omega_i \neq 0. \quad (14)$$

Since the gradient of a globally defined scalar potential must have zero circulation along every closed loop, the natural-frequency term is a topological obstruction to a global potential on $\mathbb{T}^d$.

Interaction energy. Although the full dynamics do not admit a globally defined potential in the presence of natural frequencies, the trigonometric interaction term admits a natural energy under symmetric coupling. Assume that the coupling matrix is fixed and symmetric:

$$c_{ij} = c_{ji}. \quad (15)$$

Define the interaction energy

$$E_{\text{int}}(\Theta) = -\frac{1}{2} \sum_{i,j} c_{ij} \sin \theta_i \sin \theta_j. \quad (16)$$

Then

$$\frac{\partial E_{\text{int}}}{\partial \theta_i} = -\cos \theta_i \sum_j c_{ij} \sin \theta_j. \quad (17)$$

Therefore, the trigonometric interaction term can be written as

$$\cos \theta_i \sum_j c_{ij} \sin \theta_j = -\frac{\partial E_{\text{int}}}{\partial \theta_i}. \quad (18)$$

Substituting this into Eq. (9), the full dynamics can be decomposed as

$$\dot{\theta}_i = \omega_i - \gamma \frac{\partial E_{\text{int}}}{\partial \theta_i}. \quad (19)$$

Thus, the system is a drift-plus-gradient system rather than a pure gradient flow.

The same structure can also be expressed geometrically on the embedded circle. Let

$$x_i = \begin{pmatrix} \cos \theta_i \\ \sin \theta_i \end{pmatrix} \in S^1, \quad T_i = \frac{\partial x_i}{\partial \theta_i} = \begin{pmatrix} -\sin \theta_i \\ \cos \theta_i \end{pmatrix}. \quad (20)$$

Here T_i is the unit tangent vector at x_i , satisfying $x_i^\top T_i = 0$. Since $\dot{x}_i = T_i \dot{\theta}_i$, Eq. (19) becomes

$$\dot{x}_i = \omega_i T_i - \gamma \text{grad}_{x_i}^{S^1} E_{\text{int}}. \quad (21)$$

The first term is the natural-frequency drift along the circle, while the second term is the Riemannian gradient flow induced by the interaction energy.

Why full dynamics does not admit a Lyapunov function. Although E_{int} captures the interaction part of the dynamics, it is not in general a Lyapunov function for the full system when $\omega \neq 0$. Indeed, along trajectories of Eq. (19),

$$\frac{dE_{\text{int}}}{dt} = \sum_i \frac{\partial E_{\text{int}}}{\partial \theta_i} \dot{\theta}_i \quad (22)$$

$$= \sum_i \omega_i \frac{\partial E_{\text{int}}}{\partial \theta_i} - \gamma \sum_i \left(\frac{\partial E_{\text{int}}}{\partial \theta_i} \right)^2. \quad (23)$$

Equivalently,

$$\frac{dE_{\text{int}}}{dt} = \langle \nabla_{\Theta} E_{\text{int}}, \omega \rangle - \gamma \|\nabla_{\Theta} E_{\text{int}}\|_2^2. \quad (24)$$

The second term is non-positive, but the first term is not sign-definite. Consequently, E_{int} is not guaranteed to decrease along the full dynamics, and therefore it is not a Lyapunov function for the system with nonzero natural frequencies.

Lyapunov structure in the zero-frequency symmetric-coupling regime. If the natural-frequency term is removed, *i.e.*,

$$\omega_i = 0 \quad \text{for all } i, \quad (25)$$

and the coupling matrix is fixed and symmetric, then Eq. (19) reduces to

$$\dot{\theta}_i = -\gamma \frac{\partial E_{\text{int}}}{\partial \theta_i}. \quad (26)$$

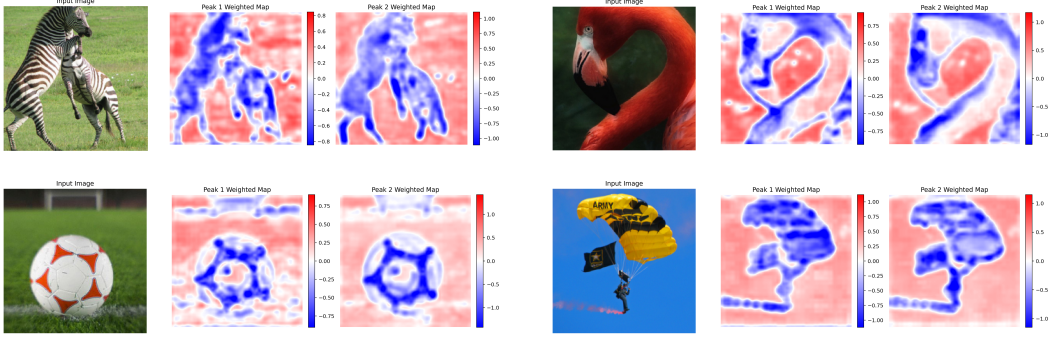


Figure 6: Additional qualitative visualizations of two-peak distributions on image recognition.

Thus the dynamics are a gradient flow on the torus:

$$\dot{\Theta} = -\gamma \nabla_{\Theta} E_{\text{int}}(\Theta). \quad (27)$$

In this case,

$$\frac{dE_{\text{int}}}{dt} = \sum_i \frac{\partial E_{\text{int}}}{\partial \theta_i} \dot{\theta}_i \quad (28)$$

$$= -\gamma \sum_i \left(\frac{\partial E_{\text{int}}}{\partial \theta_i} \right)^2 \quad (29)$$

$$\leq 0. \quad (30)$$

Therefore, in the zero-frequency symmetric-coupling regime, E_{int} is a Lyapunov function for the trigonometric interaction dynamics. In particular, isolated local minima of E_{int} correspond to Lyapunov-stable synchronized states.

Implication for WONN. The full WONN architecture contains frequency states, discrete-time updates, layer transitions and non-symmetric or state-dependent interactions. Therefore, the Lyapunov structure above should not be interpreted as a global stability theorem for the full model. Nonetheless, E_{int} provides a principled interaction energy for the trigonometric Winfree core and serves as a practical diagnostic of phase alignment. In Maze-hard (Sec. 3.3), we use this diagnostic for energy-based test-time selection among multiple stochastic phase trajectories.

E Additional Visualization

E.1 Additional Visualization on Image Recognition

Here we provide additional qualitative visualizations for image recognition. As shown in Fig. 6, these examples exhibit similar patterns to those discussed in Sec. 3.2, further supporting the observation that the phase modes of WONN capture complementary visual structures.

In addition to the static weighted maps shown in Sec. 3.2, we further visualize their layer-wise evolution in Fig. 7. The figures show how the weighted maps associated with the two dominant phase peaks evolve across recurrent Winfree dynamics steps and network depth. We observe that the phase-weighted responses are repeatedly refreshed at each layer: early layers mainly capture weak and local visual structures, while deeper layers progressively organize these responses into more coherent global semantic patterns. This behavior suggests that the recurrent synchronization dynamics of WONN gradually refine and reorganize visual representations over depth and time.

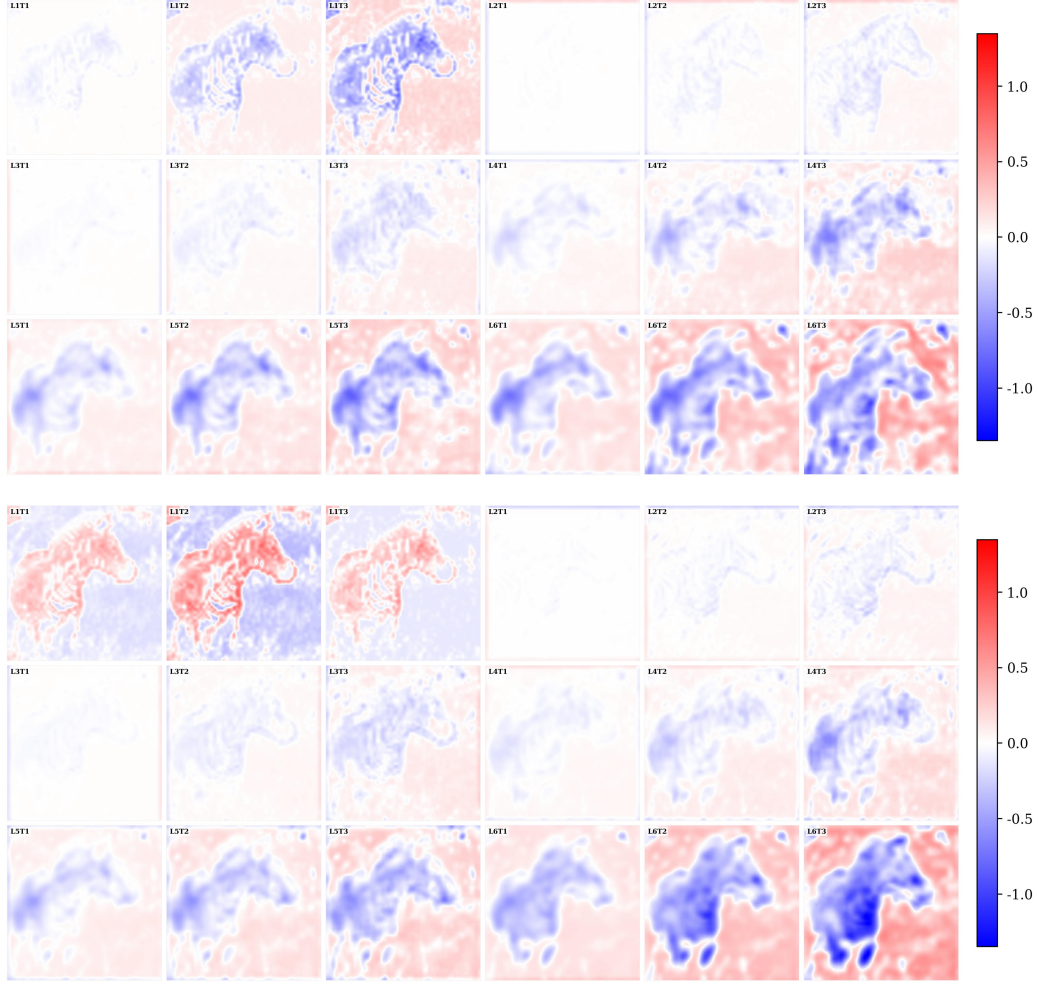


Figure 7: Layer-wise evolution of the weighted maps associated with the two dominant phase peaks in *WNN* on image recognition. **Top**: weighted map corresponding to the first dominant phase peak. **Bottom**: weighted map corresponding to the second dominant phase peak. Here L denotes the layer index and T denotes the Winfree dynamics step within that layer. Panels are arranged according to the actual forward trajectory, from $L1T1$ to $L6T3$. Across layers, the phase-weighted responses are progressively refreshed and reorganized, evolving from weak local activations toward more coherent global semantic structures through the recurrent synchronization dynamics.

E.2 Additional Visualization on Maze-hard

This subsection presents additional Maze-hard examples for qualitative analysis.

Fig. 8 illustrates the evolution of accuracy and interaction energy during training on the Maze-hard task. While the energy initially increases in the early stage of optimization, it subsequently decreases as training progresses, accompanied by a steady improvement in accuracy. This inverse correlation suggests that lower interaction energy is associated with more accurate solution states, indicating that the energy provides a useful proxy for solution quality in the learned synchronization dynamics.

Fig. 10 shows the complete temporal evolution of the example discussed in Sec. 3.3, while Fig. 11 provides several additional instances. In the path prediction maps, black cells represent walls, red and green cells denote the start and goal, respectively, blue cells indicate the predicted path, and purple cells correspond to invalid wall-crossing predictions. Across different examples, we observe a consistent synchronization-driven dynamics: during the early stages, the network activates multiple candidate path fragments simultaneously rather than directly converging to a single solution. As the oscillatory dynamics evolve, these fragments progressively synchronize, compete, and merge into coherent global trajectories, while inconsistent or invalid paths are gradually suppressed. Eventually, the dynamics converge to the shortest valid path connecting the start and goal. These visualizations suggest that synchronization dynamics provide a very natural mechanism for coordinating distributed local candidates into globally consistent reasoning solutions.

Fig. 9 show the temporal evolution of *HRM* on the same example discussed in Sec. 3.3, including both path predictions and probability maps. We visualize the first 12 H-block updates and regard them as the first 12 evolution steps. The results suggest that *HRM* remains largely inactive during the early stages, then begins to generate irregular predictions around $T = 4$, and subsequently undergoes an abrupt transition around $T = 6$ that recovers most of the correct path. The remaining steps only introduce minor refinements. This abrupt, insight-like behavior contrasts sharply with the progressive path formation exhibited by *WONN*.

E.3 Additional Visualization on Sudoku

This section provides qualitative visualizations on the Sudoku task. Fig. 12 shows that *WONN* follows a dynamical solving process similar to that observed on Maze-hard. At early time steps, the model tends to perform a global synchronized exploration of possible assignments. During subsequent evolution, these predictions are gradually refined, and the model progressively converges to the correct solution.

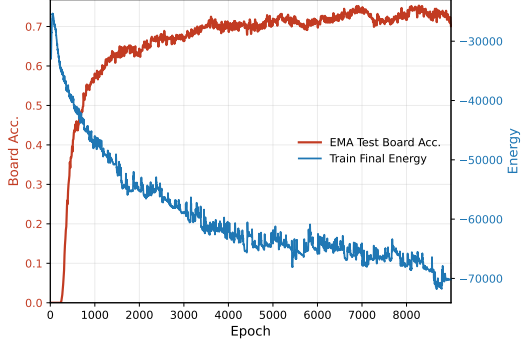


Figure 8: Accuracy–energy dynamics on Maze-hard. We report the test accuracy and the corresponding energy over training epochs. After a slight increase in the early stage, the energy gradually decreases as training proceeds, whereas the accuracy exhibits a steady improvement trend.

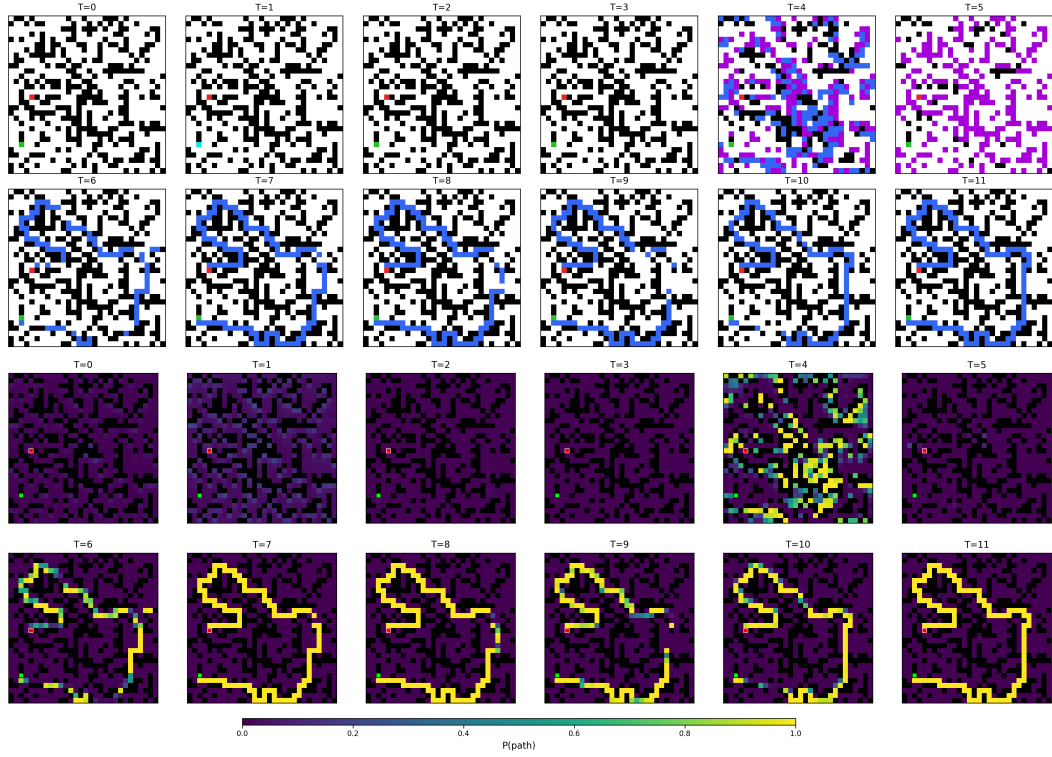


Figure 9: Partial temporal evolution of *HRM* on the example shown in Sec. 3.3. **Top:** discrete path predictions over time. **Bottom:** predicted path probability heatmaps over time. *HRM* remains mostly inactive during early H-block updates and then undergoes an abrupt, insight-like transition, while *WONN* progressively synchronizes diffuse candidate path fragments into a coherent valid path.

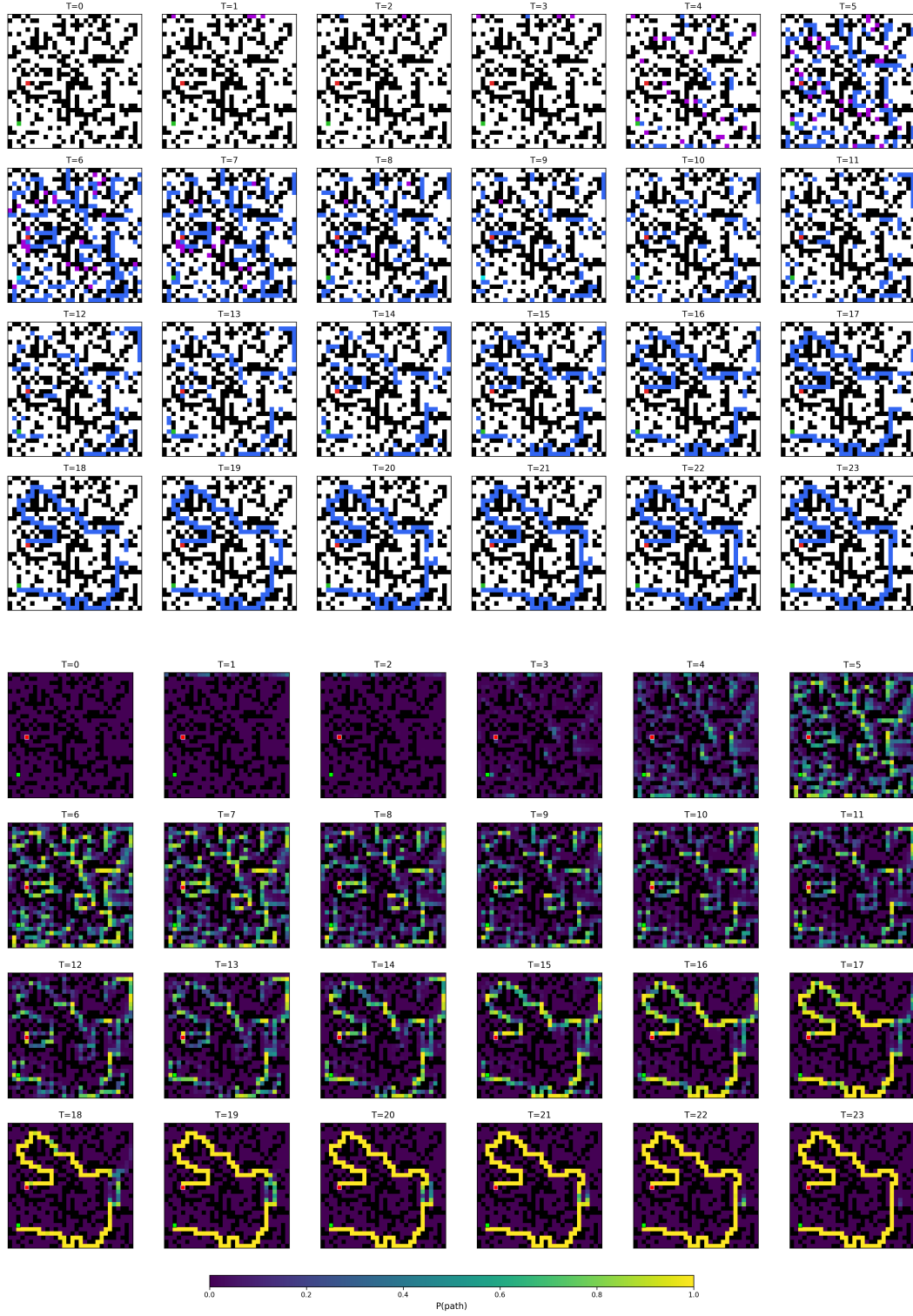
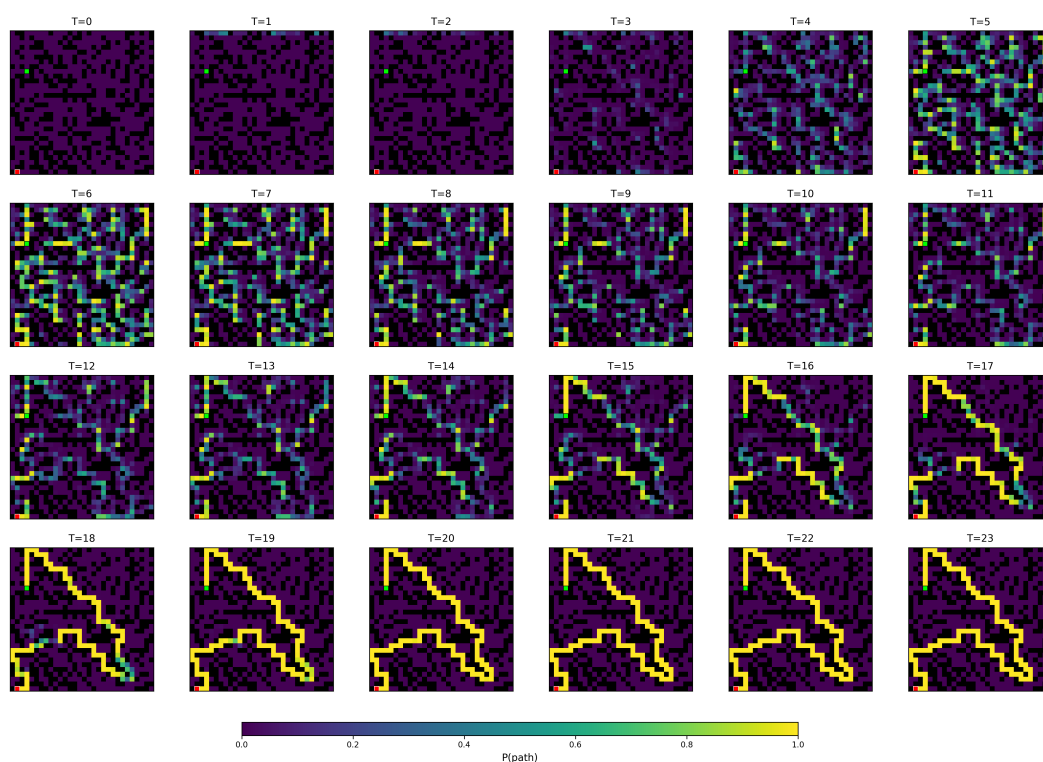
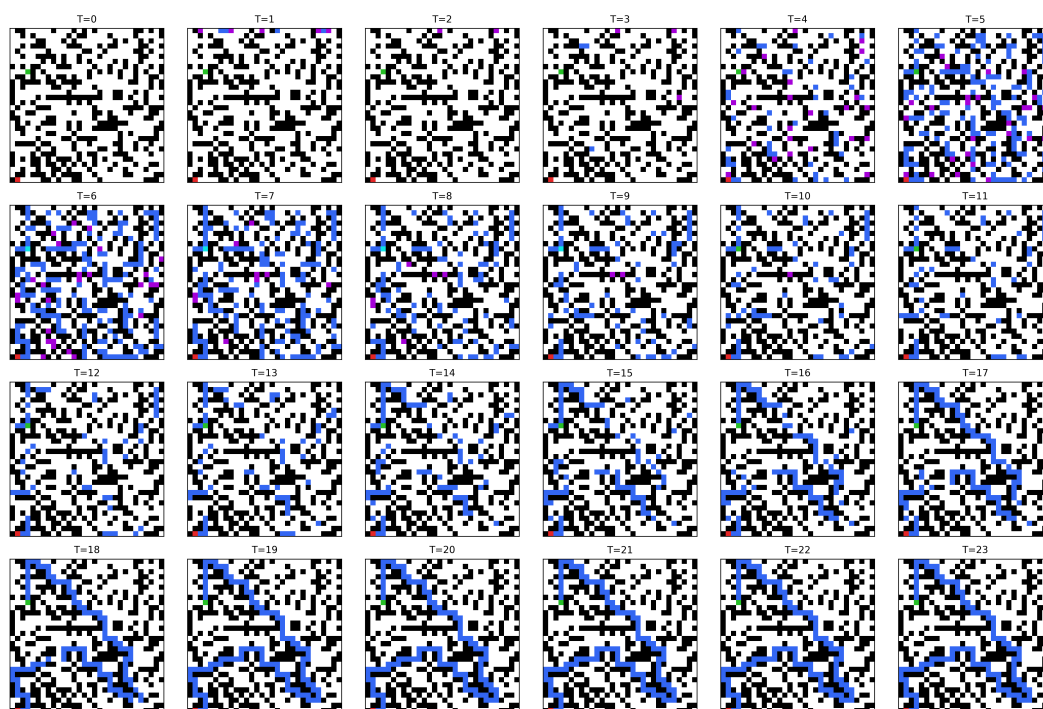
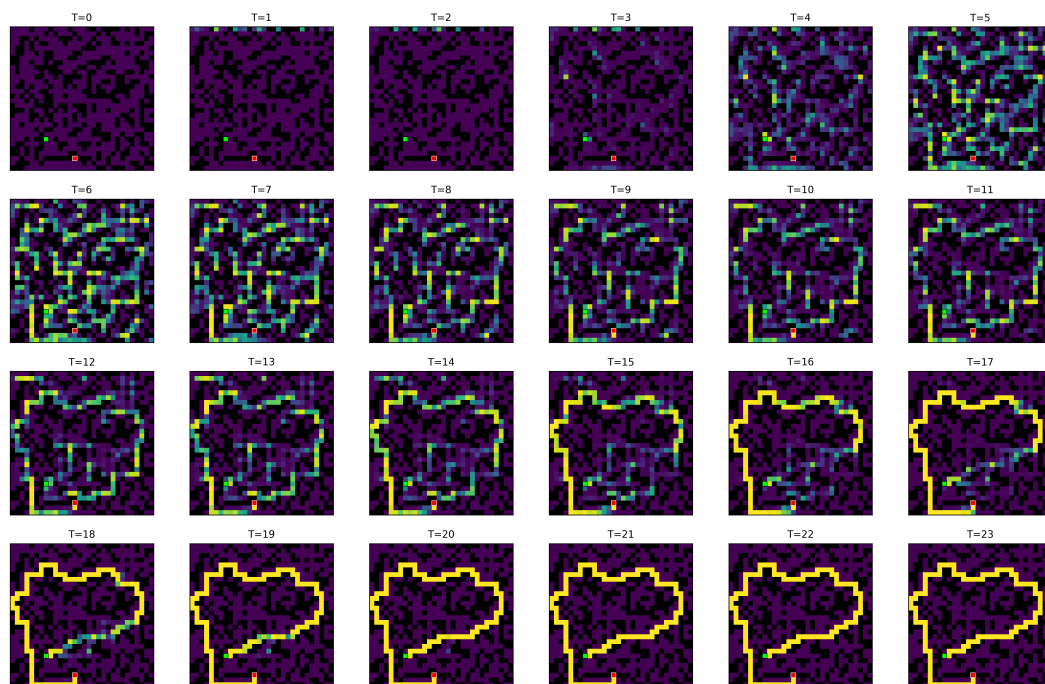
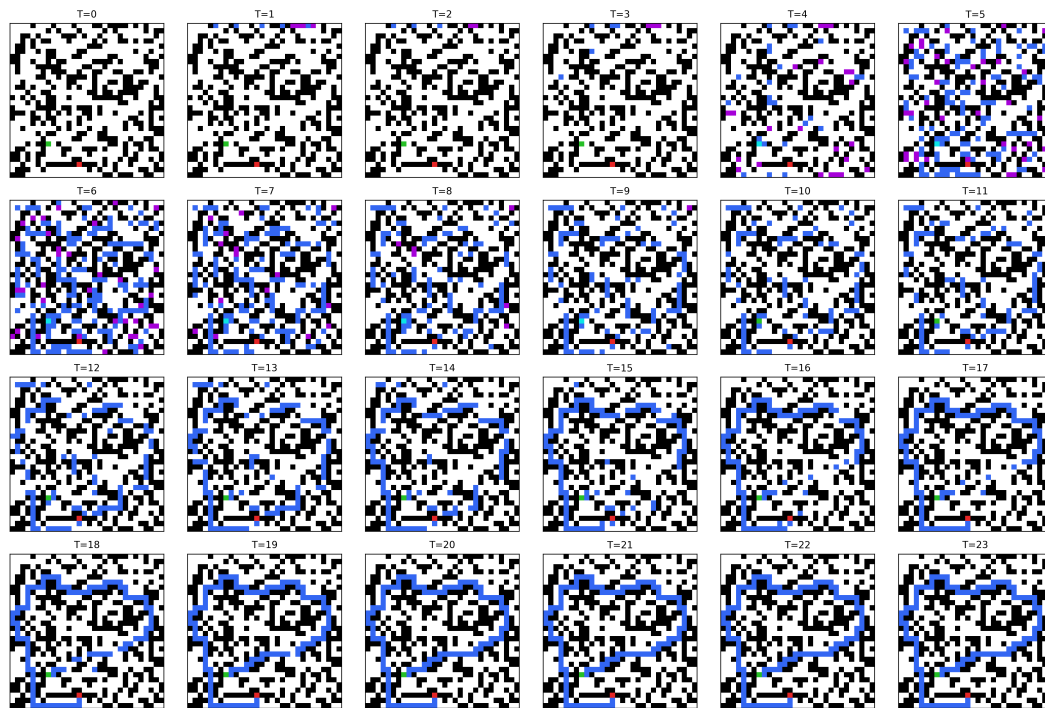


Figure 10: Complete temporal evolution of the example shown in Sec. 3.3. **Top:** discrete path predictions over time. **Bottom:** predicted path probability heatmaps over time.





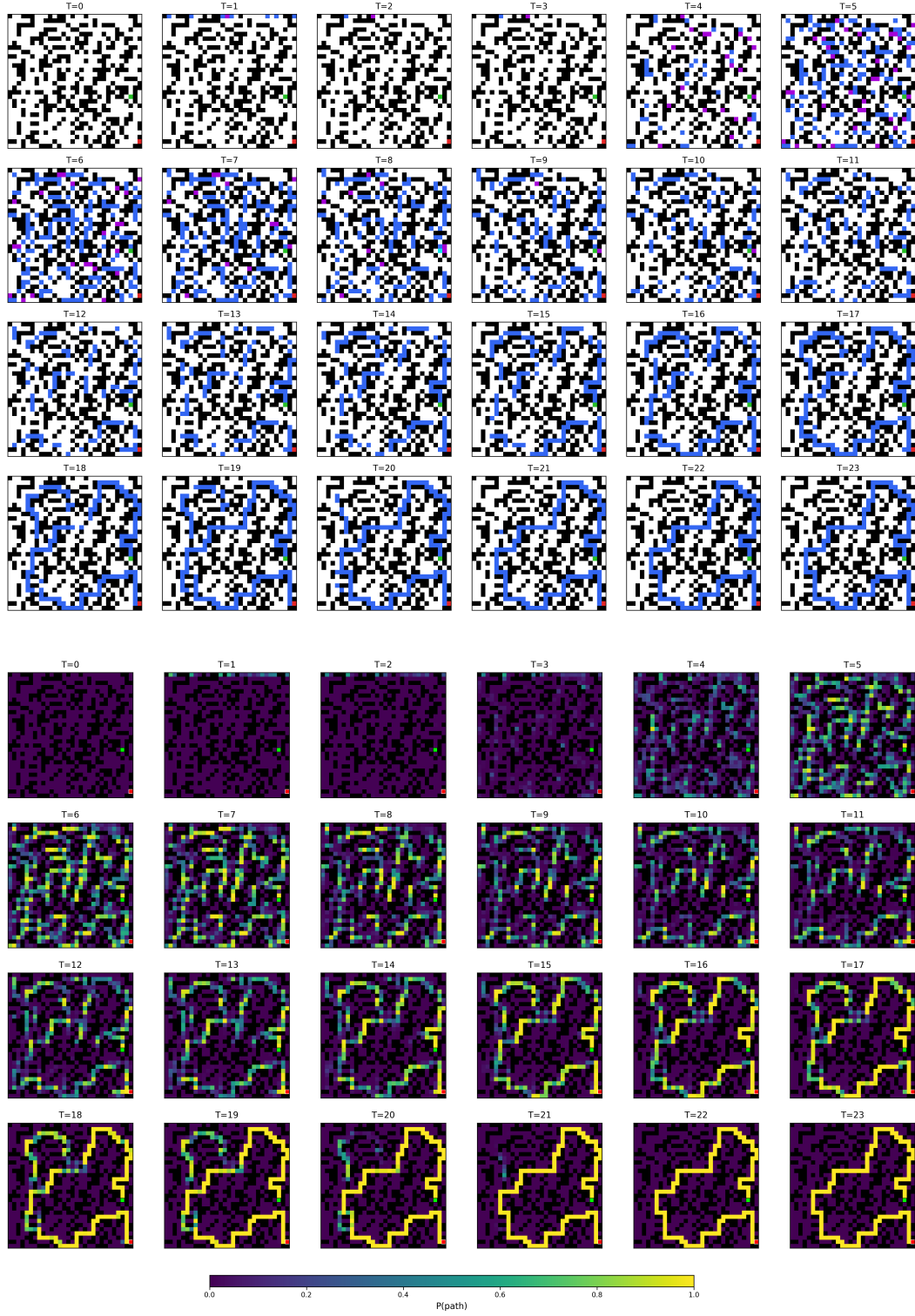


Figure 11: More Maze-hard pathfinding examples. Each pair shows the discrete path prediction and the corresponding path probability heatmap for one maze instance.

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Figure 12: Temporal evolution of *WONN* on Sudoku. Early steps show global synchronized exploration over candidate digit assignments, while later steps progressively refine these predictions and converge to a correct globally consistent solution.